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A systematic scoping literature review into temperate food forests: an overview of the current knowledge and a research agenda

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Abstract Food forestry might be a sustainable complement to agricultural systems in the temperate biome. These systems need to address ongoing challenges, such as biodiversity decline, water quality issues, and greenhouse gas emissions. However, while examples of temperate food forestry exist, a comprehensive overview of literature is lacking, limiting insight into how to grow and maintain these forests, as well as the potential of these forests in addressing environmental and social challenges. This study conducts a systematic scoping literature review to collate the existing knowledge on temperate food forestry. Topics addressed in this paper include the challenges and benefits of food forestry, the different phases of establishment, and the integration of Indigenous knowledge. The paper concludes with a research agenda, highlighting the need for more research into mature food forests, including harvesting practices and revenue models; more research in temperate climates in the southern hemisphere, including South America, Australia and New Zealand; and a stronger focus on the integration of Indigenous knowledge in food forestry to address agricultural challenges.

Keywords Food forestry · Forest gardens · Agroecology · Permaculture

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Introduction

The incorporation of trees into farm systems has been practiced worldwide for over 6000 years (Gordon et al. 2018). Historical records demonstrate the widespread adoption of tree-based farming practices for the provision of food, medicine, and fuelwood. Examples include the cultivation of crops and livestock in the Netherlands in 2500 BC, where farmers combined grazing and cropping with the existing forest system (Oosterbaan and Kuiters 2008). In central America, farmers have traditionally mimicked natural forest structures to optimise food production

for millennia by cultivating multiple layers of vegetation, including tall coconut trees, mid-level banana trees and shrub layers of cacao or coffee (Gachuiri et al. 2022; Wilken 1978). In Asia, specifically the Philippines, the Indigenous population has practiced shifting cultivation where some trees were deliberately retained when clearing forest land. Those trees were subsequently used in rice fields, where they provided shade and contributed to retaining moisture to help mature the grain (King 1987). Another example includes homegarden systems, which are often multistorey combinations of a range of trees and crops, which can be grown in association with domestic animals. These homegardens are often grown around the homestead and one of the oldest forms of agroforestry in countries like Indonesia, Sri Lanka and Bangladesh (Sharma et al. 2022).

Although societies worldwide have practiced the integration of trees into the farming landscape at different points in history, the term 'agroforestry' did not emerge until after the Second World War (Nair et al. 2021a, b). The green revolution of the 1960s promoted high-yielding crop varieties, often grown in monocultures, the intensive use of fertiliser and pesticides, and large-scale irrigation. This increased production significantly, especially in the Global South. But environmental failures associated with this intensive land use called for a need for more sustainable, integrated practices (Aguilar-Rivera et al. 2020). This need did not only derive from an environmental perspective, but especially in the Global South also from a social perspective, where resource-poor farmers found themselves unable to farm. At the same time, advancements were made in commercial tree growing. This saw agroforestry being promoted as a way to combine trees and crops on the same land unit, supporting both tropical and temperate regions in addressing ecological degradation. Since the 1970s, research on agroforestry has had multiple foci (Noordwijk et al. 2019). In the 1980s and 1990s there was a dominant emphasis on soil fertility restoration, particularly in tropical regions (Sanchez 1999). From the 1990s onwards, there has been an increasing focus on the food-producing role of agroforestry, driven by efforts to improve the quality and availability of Indigenous food products (Rolo 2022). More recently, the term 'land maxing' has gained traction, especially in tropical and subtropical agroecosystems, which combines domestication and commercialisation of Indigenous food trees with land restoration, thus strengthening local communities, food sovereignty, and the economy (Leakey and Harding 2025).

Currently, agroforestry is defined by the FAO as: 'a collective name for land-use systems and technologies where woody perennials (such as trees, shrubs, palms, bamboos, etc.) are deliberately used on the same land-management units as agricultural crops and/or animals, in some form of spatial arrangement or temporal sequence' (FAO 2025). Many examples of agroforestry derive from tropical countries, with small farms and systems that are less industrial than in the Global North (Woodhill et al. 2020). Current estimations, which assume land is in agroforestry when it meets 10% canopy cover in agricultural areas, shows that 46% of the world's agricultural area is in agroforestry, and 80% of this land is located in tropical areas in Southeast Asia, Central America, and South America (Zomer et al. 2009). This suggests that only 20% of agroforestry land is located in temperate zones, but research points out the greater potential for agroforestry in the temperate zone due to climate mitigation and adaptation benefits, biodiversity enhancement, and erosion reduction (Hirth et al. 2025; Knook et al. 2025; Wolz et al. 2018; Park et al. 2019; Wartman et al. 2018). In this paper, we refer to temperate agroforestry as agroforestry in the two wide bands of latitude between roughly 23.5 and 66.5 degrees north and south of the equator. These areas have been characterised by moderate climates with clear seasonal changes, avoiding extreme heat and extreme cold (IPCC 2022). Agroforestry systems in such climates must be better adapted to seasonal variations with colder winters and frosts, as well as warmer summers and periods of drought, than those in tropical and sub-tropical regions.

Forms of temperate agroforestry

Agroforestry comes in a range of forms, but all forms intentionally integrate trees with crops and/or livestock to optimise ecological and economic benefits. Nair et al. (2021a, b) provide an overview of different classifications of agroforestry systems, therein stating that agroforestry practices in temperate regions are less diverse and complex than tropical agroforestry systems. The Association for Temperate Agroforestry (2025) has adopted a classification of the following practices for temperate agroforestry: 1)

alley cropping, which refers to trees planted in single or grouped rows with crops grown in the (generally wide) spaces in between; 2) forest farming, which refers to the production of specialty crops (e.g. for medicinal use) under existing forest; 3) silvopasture, in which trees are combined with pasture on which livestock grazes; 4) riparian buffer, where strips of perennial vegetation is planted between pasture or crops and waterways; 5) windbreaks, where rows of trees are planted around arable or pastoral land to protect crops or livestock; and 6) (urban) food forests, where multilayered, perennial systems are designed to mimic natural forest ecosystems by integrating diverse edible plants. The sixth category, (urban) food forestry, has only been a recent addition to this classification, with the 2018 edition of Temperate Agroforestry Systems noting that these food forests are generally not considered an important form of agroforestry in temperate zones (Gordon et al. 2018).

However, more recent studies (Albrecht and Wiek 2021a; Nytofte and Henriksen 2019) point out the potential of these food forests by highlighting the opportunities to provide local food, enhance biodiversity, and support community wellbeing. More knowledge of food forestry in the temperature zone is required, with Albrecht and Wiek (2021a) calling for a need to study the financial feasibility of food forests, as well as highlighting challenges such as labour-intensive harvesting practices and high land prices. Bukowski and Munsell (2018) add that most of the researched temperate food forests have been established win the last five years, so add to the call for more literature on the maintenance of these forests after establishment.

Food forestry

To contribute to the call for more knowledge on temperate food forestry, this study presents a literature review into temperate food forestry (from here on referred to as 'food forestry'). This review aims to: i) synthesise the current state of the literature; ii) determine what this tells us about the main research themes in the temperate food forestry literature; and iii) identify knowledge tensions and gaps to direct future research into this area. It helps identify the existing knowledge, as well as knowledge gaps to steer the future research agenda in this field.

But first, what is temperate food forestry? In peer-reviewed scientific literature, the first studies that use the terminology 'food forestry' in temperate zones were published in 2004 (Wiersum 2004). Since then, the practice has spread with food forests being planted in both rural and urban settings in temperate countries, including (though not limited to): Germany, the Netherlands, Portugal, the United Kingdom, and the United States of America (Albrecht and Wiek 2021b; Castro et al. 2018; Pilgrim et al. 2018; Roodhof 2024).

There is no universal or widely accepted definition of temperate food forestry, but a combination of key characteristics were identified to guide our interpretation and understanding, as well as our search terms. These characteristics are: 1) the inclusion of edible species; 2) a polyculture design; and 3) the inclusion of perennial species. In relation to the inclusion of edible species, Albrecht and Wiek (2021a) state that food forests and forest gardens are "both defined as multi-strata ecosystems using mostly edible, perennial plants" (pg. 92). Hemmelgarn and Munsell (2021) agree, and suggest an additional term—"forest garden"—to be more inclusive of wider food goods and services provided by these systems. A second key characteristic is the polyculture element of food forests to include many species (Albrecht and Wiek 2021a; Pilgrim et al. 2018). These different species are then commonly distributed over different layers such as canopy, understory, ground cover, and roots and climbers, using words like multistrata, multi-layered, and multistorey to describe the structure. This emulates the importance of mimicking the natural forest, or forest-like, systems (Albrecht and Wiek 2021a; Hemmelgarn and Munsell 2021). A third characteristic is the presence of perennial species in these systems (Armstrong et al. 2021; Stoltz and Schaffer 2018). Few studies define a food forest by its size, though they are referenced as being notably smaller than other land use types like commercial farming or fruit growing (Albrecht and Wiek 2021a; Pilgrim et al. 2018). While authors such as Kowalski and Conway (2019) and Cariñanos et al. (2022) use the word urban to differentiate the location of the food forest itself, their definitions are otherwise comparable. Therefore, our interpretation of the concept of temperate food forestry is based on the above characteristics: it refers to the intentional design and management of forest-like, multi-layered systems with a

primary focus on growing edible, perennial plants within the temperate biome.

Food forestry and related concepts

Before we outline the methodology of our study (including our search terms), we see it as important to place food forestry in relation to other key concepts, such as 'agroecology', 'permaculture', 'regenerative agriculture', 'orchards', and 'food or home gardens'. This will increase understanding into why we have looked into food forestry specifically as a stand-alone area of study.

Agroecology is an integrated approach that applies ecological concepts and principles to the design and management of agricultural systems. It emphasises biodiversity, local knowledge and social equity to enhance food security and nutrition (HLPE 2019). The 2019 report by the High Level Panel of Experts on Food Security and Nutrition follows Gliessman (2014) by defining five levels of agroecology: (1) increasing efficiency of industrial inputs, for example, by reducing pesticide use; (2) substituting synthetic inputs with biological alternatives, for example, through the use of organic fertilisers; (3) redesigning farming systems based on ecological processes and diversification, which can include agroforestry techniques; (4) reconnecting food producers and consumers through local markets and short supply chains; and (5) transforming food systems towards sustainability and social justice, by addressing power relations and governance. Temperate food forestry mostly fits into level three, as it redesigns agricultural systems to mimic natural forest ecosystems (Albrecht and Wiek 2021b). It also contributes to levels four and five, with the potential for food grown in food forests to be sold into local markets, and being closely tied into Indigenous knowledge. Hence, while the two concepts are not synonymous, food forestry can be placed within the larger umbrella of agroecology.

Another key concept often linked to food forestry is permaculture. Permaculture mimics, similar to food forestry, the patterns and relationships found in natural ecosystems (Ferguson and Lovell 2014). Permaculture goes far beyond solely food forestry, and covers several other approaches, such as permaculture gardens (diverse vegetable and herb gardens using companion planting) and agroforestry systems more widely. In food forestry specifically, the similarity to permaculture lies in the creation of systems that mimic natural ecosystems and are self-sustaining and low-input.

Regenerative agriculture has been compared to agroforestry (Elevitch et al. 2018), more specifically to silvopastoral systems, where both livestock and trees are present. While linkages to food forestry are limited, they can be observed through the core environmental concern that regenerative agriculture tries to improve overall ecosystem health. While regenerative agriculture is not a new term, nor is the spirit of it, it has received increased attention in the last 10 years (Giller et al. 2021). Similarly to food forestry, there is no commonly accepted definition of regenerative agriculture. A common perspective is to view it as providing an alternative to, and/or challenging the dominant agricultural system of, high input, intensively farmed monocultures (LaCanne and Lundgren 2018). As a result, further similarities can be observed through the commonly shared principles of increasing diversity of plant species, favouring perennial plants over annuals, and improving the health of the natural environment—all with a strong focus on soil improvement (LaCanne and Lundgren 2018; Rhodes 2017). Though the two systems are not synonymous, many similarities exist between the two concepts.

Orchards are often understood as a piece of enclosed land, intentionally planted with neatly planted fruit and/or nut trees (Williamson 2020). While in the modern day these are often the same tree species planted in uniform, high-density rows with narrow canopies, traditional orchards in the 17th and 18th Century were often much smaller, less intensively managed, designed for aesthetic purposes, and contained open-grown trees (Biodiversity and Information Research Group, 2016). For these reasons, parallels can be drawn between historical orchards and the food forests we know today. However, we urge caution when comparing food forestry with contemporary commercial orchards (and their focus on high density, intensively managed monocultures) (Noble 2021). Even the closest comparison—community orchards—differ from food forests in the sense that food forests are deliberately set up in patterns and layers that mimic a natural forest's ecosystem (Bukowski and Munsell 2018), where orchards lack this configuration (Lovell et al. 2021).

Food gardens, also called home gardens by Gordon et al. (2018), are particularly common in the tropical and subtropical biomes. They are generally small-scale and intensively managed near homes or communities (Poveda 2016). Research in the tropical biome has found that food gardens have higher biodiversity than large-scale commercial land-use systems (Kumar and Nair 2004). Furthermore, beneficial plant-to-plant interactions reduces the requirement for external inputs such as fertiliser and pesticides (Park and Higgs 2018). With the terms 'home gardens' and 'food gardens' used generally to indicate tropical food forestry (Nair et al. 2021a, b), our study on 'food forestry' is focused on those in the temperate zone.

Methodology

Literature reviews play a critical role in research, not only by identifying and justifying research gaps, but also to avoid unnecessary duplication of effort. They serve as valuable tools to identify theories, frameworks, and trends within a specific domain, thereby guiding both current and future research (Page et al. 2021). Scoping reviews are informed by a systematic approach of searching databases and extracting data. Their focus allows the identification of all available evidence on a particular topic, key concepts and characteristics, and knowledge gaps (Munn et al. 2018). For this study, a systematic scoping review was identified as an appropriate method to synthesise the existing evidence on temperate food forestry and provide a transparent and rigorous basis for identifying research priorities (Munn et al. 2018). A similar approach to the systematic literature review in the area of agroforestry has been taken by Tranchina et al. (2024), which confirmed the appropriateness of this approach for our study.

Applying the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRIMSA) protocol (Moher et al. 2009) in our systematic scoping review enables an exploration of the temperate food forestry literature, thereby summarising the literature on food forestry in temperate climates, as well as identifying gaps in our understanding of environmental, economic, and socio-cultural perspectives.

Protocol

To conduct the literature review, we utilised the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) protocol by Moher et al. (2009) (see Fig. 1). This approach provides a framework for systematically searching the literature and has been applied in a range of disciplines (Page et al. 2021).

Search engines and terms

Our search was conducted using the Scopus and Web of Science databases and the following search terms were used in this review. Based on the critical review of literature related to food forestry (Sect. "Food forestry and related concepts"), and a scan of similar concepts, both food forest and food garden were included as search teams; although food forestry is generally used in temperate systems (Albrecht and Wiek 2021b; Poveda 2016). We followed Boolean operators (AND, OR, "") and used truncations (*). The three search strings that were entered into both Web of Science and Scopus were:

(1) "food forest*" AND "temperate"
(2) "food forest*"
(3) "forest* garden*", "home garden*" OR "edible forest*"

To reduce the number of results, while still maintaining robustness, the search was restricted to journal articles and reviews only. We conducted the search in June and July 2024 and the date was not restricted to ensure all possible articles were captured. The article language was restricted to English. This scan resulted in 1,220 records with 683 duplicates. Removing duplicates reduced our records to 537.

Criteria for inclusion and exclusion

Search results mentioning "tropical", or studies that did not focus on the temperate biome, were excluded from the study, along with forest farming (the farming of trees, which does not follow the characteristics of food forestry). This decreased the number of potentially relevant records to 414. Full abstract screening then occurred, following the inclusion criteria of

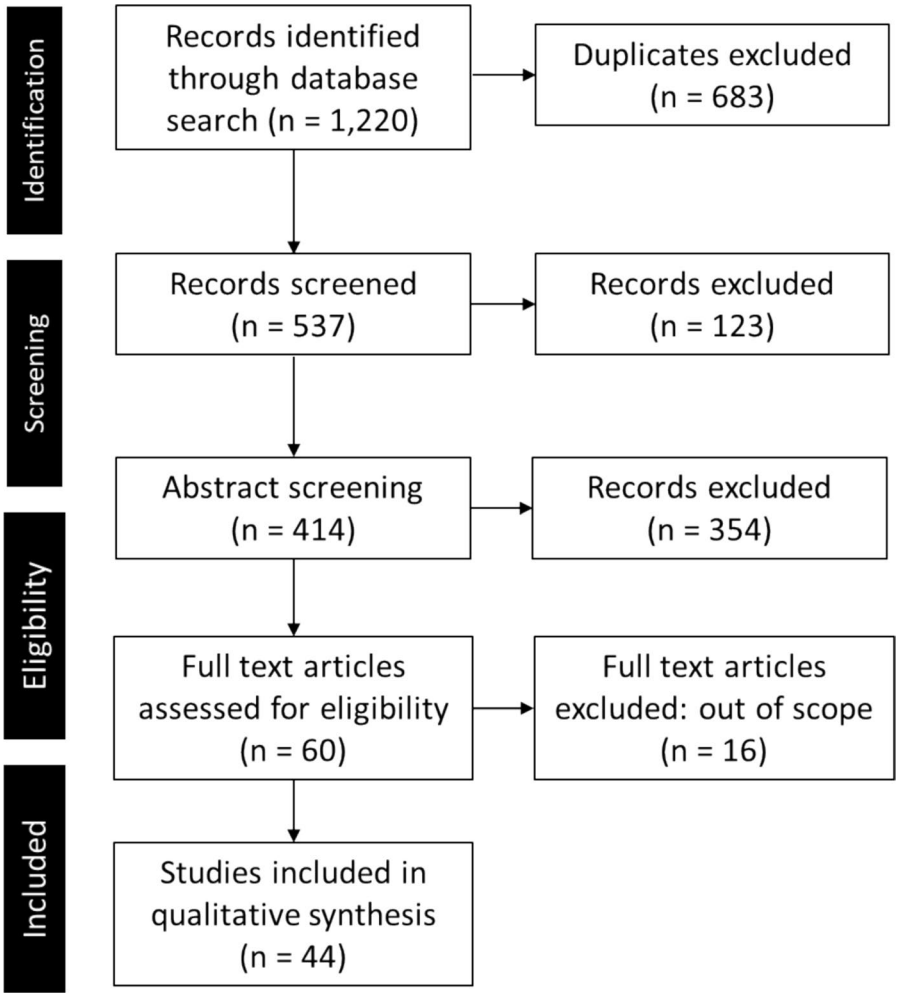

Fig. 1 PRISMA diagram based on Moher et al. (2009)

studies actively referring to managed food forests and forest gardens. This resulted in 59 records.

The references of the included papers were reviewed to ensure the inclusion of all relevant papers on the topic. Snowballing enabled us to identify literature relevant to the overall topic of food forestry (for example, an overview of food forestry in the tropical zone of Brazil (Vannozzi Brito and Borelli 2020), and relevant book chapters in relation to food and agroforestry (Toensmeier 2017; Young 2017)), but did not result in further articles being included in this review.

Selection of the final studies

Applying our inclusion and exclusion criteria to titles and abstracts showed articles with apparent relevance to our overall study, but to identify the final set of results, the full texts of all 59 articles were read. Through this process, the final set of 44 studies was identified for inclusion in this study.

The 16 articles that were excluded in the last phase of the selection process did not meet the inclusion criteria for several reasons. For example, two articles were excluded, as they turned out to be opinion pieces. Other articles used the terminology 'food' and 'forestry', but when reading these articles in whole, they did not discuss temperate food forestry, instead focusing on foraging in wild forests. The thorough reading of the articles in full thus led to the change from 59 to 44 included studies.

Extracting data for analysis

Using Microsoft Excel, a data extraction form was developed. This included the names of authors, year of publication, journal titles, primary country(ies) of study, keywords, research methods, and preliminary themes. Our analysis, which was explorative and inductive, meant that the themes arose organically from the studies.

Findings

Food forestry in the literature—descriptive analysis

The oldest publication included in this review was from 2004. The number of publications increased from 2013 with a peak in 2019. Only one study was published the following year, perhaps coinciding with the coronavirus (COVID-19) pandemic, but publications returned to an average of about five per year from 2021 to 2024. This illustrates that temperate food forestry is still a relatively novel area of research, with studies in this area steadily increasing.

Most of the studies within our search criteria were conducted in the United States of America (31%), Europe (24%), Canada (21%), and the United Kingdom (17%). Within Europe, Sweden was the most common country context studied (mentioned in 10% of the total publications), followed by Italy (7%), the Netherlands (7%), Germany (5%), and Portugal (2%). There were no examples of food forests from Australia and only one from Aotearoa New Zealand, presenting another area of future research worthy of investigation. Furthermore, there were no articles focusing on food forestry in temperate regions in South America, only in tropical areas such as central Brazil.

Growing a temperate food forest

Of the 44 studies reviewed, 17 specifically focused on outlining the processes involved in creating food forests, and the relative success factors of doing so. Perhaps because so many food forests are young (Albrecht and Wiek 2021a), the majority of the studies included in this review (14/17 studies) is focused on the planning, design, funding, site choice, and networks required to initiate a food forest project. Some research explores the establishment phase, where a site is prepared, planted, and managed most intensively. Relatively few articles investigate processes involved in ongoing management of mature food forests, indicating a research gap.

The initiation phase: pre-planting

In the initiation phase the following aspects are key: (i) the policy environment; (ii) a comprehensive entrepreneurial system; and (iii) careful planning and site design (Albrecht and Wiek 2021b; Castro et al. 2018; Clark and Nicholas 2013; Hemmelgarn and Munsell 2021; Kowalski and Conway 2023; Pilgrim et al. 2018; Schafer et al. 2019; Wiek and Albrecht 2022; Yen and Chen 2016). The policy environment, is particularly important in peri-urban and urban areas where land use regulations can be incompatible with food forestry and urban agriculture (Albrecht and Wiek 2021b; Castro et al. 2018; Hemmelgarn and Munsell 2021; Kowalski and Conway 2023). Changes in policy have been recorded. For example, policies in Seattle in the United States of America were changed to support tree cover and urban agriculture (Albrecht and Wiek 2021b). In the Netherlands, food forest establishment is supported by the Green Deal, a public–private partnership that includes beneficial policies, research coordination and funding for planning and implementation of food forest projects (Albrecht and Wiek 2021b; Wiek and Albrecht 2022).

Secondly, a comprehensive entrepreneurial ecosystem is needed to provide support services to food forests, including training, financing, legal advice, and political advocacy (Wiek and Albrecht 2022). Investment is high and revenue is low during the first few years of establishing a food forest (Albrecht and Wiek 2021b). Access to personal funds is perhaps one reason why so many food forest founders are older and high-educated (Pilgrim et al. 2018; Roodhof 2024). Generally, Albrecht and Wiek (2021b) found that food forests were far more likely to succeed if founders or site managers had agricultural and entrepreneurial skills and knowledge, as well as professional equipment. Of particular importance was knowledge about working with the diversity of plants and services chosen for the food forest, as well as practical business knowledge, such as fundraising, bookkeeping, marketing, and human resources (Albrecht and Wiek

2021b). Similarly, networks were identified as critical components of early and long-term success (Albrecht and Wiek 2021b; Pilgrim et al. 2018; Roodhof 2024; Schafer et al. 2019; Wiek and Albrecht 2022; Yen and Chen 2016). Examples of entrepreneurial networks included working with local councils to foster awareness, policy support, site access (Albrecht and Wiek 2021b; Pilgrim et al. 2018), and networking with others in the food forestry community to learn from each other and enjoy collective enthusiasm (Pilgrim et al. 2018; Roodhof 2024; Yen and Chen 2016).

Lastly, careful planning and design is critical for productivity and overall success of a food forest (Albrecht and Wiek 2021b; Pilgrim et al. 2018). For example, the design of professional site plans in collaboration with permaculture experts, landscape architects, students and communities (Albrecht and Wiek 2021b; Park and Higgs 2018). There are a wide variety of plants that could be included in a temperate food forest, and over 230 specific species were mentioned in literature (e.g. Armstrong et al. 2023, 2021; Askerlund and Almers 2016; Björklund et al. 2019; Bosco and Thomas 2023; Castro et al. 2018; Clark and Nicholas 2013).

Björklund et al. (2019) point out that there is often no universal design due to the need to suit each food forest's climate, beneficiaries, and overall goals. Still, some principles emerged in the literature such as the need to focus on fewer, high-value crops in commercial harvest systems, and the need to reduce light competition by choosing shade-tolerant plants with good spacing in temperate systems (Björklund et al. 2019). Wide spacing of plants also reduces humidity, which protects against plant-fungal infections (Pilgrim et al. 2018). Furthermore, site selection is important, although this is highly dependent on affordability and appropriate land-use classifications (Albrecht and Wiek 2021b; Wiek and Albrecht 2022). Drought can be challenging, so having water rights close to or on the land is crucial for the establishment period (Albrecht and Wiek 2021b).

The establishment phase: year 0–3 of the food forest

Three studies mentioned methods of site preparation including tilling, cardboard cover, and weed-suppressing material as important throughout the establishment phase (Björklund et al. 2019; Pilgrim et al. 2018; Riolo 2019). Initial soil properties, organic matter and nutrient levels are important for successful establishment; for example, low pH impacts fruit tree establishment (Björklund et al. 2019). Affordable land for food forests is often marginal and degraded, requiring several years of soil remediation (Albrecht and Wiek 2021b). The application of green and composted mulch accelerates establishment and improves later harvests (Björklund et al. 2019). Pests, such as grazing herbivores and smothering weeds, pose some of the greatest unforeseen challenges during this phase of development (Pilgrim et al. 2018).

Maintenance poses another challenge during this phase. Björklund et al. (2019) found that farmers spent 25 h on average establishing tree and shrub layers. Pilgrim et al. (2018) found that some food forest managers had not anticipated how long it would take for trees to yield crops in large quantities. Solutions mentioned the inclusion of annual high-productivity crops around perennials to generate revenue during the establishment period (Pilgrim et al. 2018; Schafer et al. 2019). Björklund et al. (2019) also note the importance of establishing the tree and shrub layers before the herbaceous layer to avoid adversely impacting woody perennial growth; this can be a costly lesson to learn.

Overall, as mentioned by Riolo (2019), as most studied food forests are less than five years old, more research is needed to identify best design choices and look further into the environmental, health, and well-being benefits of food forestry.

The ongoing management: year 4 and beyond

Most food forests studied in literature are younger than five years old. Although no study has focused solely on the ongoing management, we did identify two factors discussed around the ongoing management mentioned in the literature: i) the management of time expectation; and ii) revenue streams.

Firstly, Pilgrim et al. (2018) found that the majority of food forest managers in the United Kingdom spent less time maintaining their system than they had anticipated, while acknowledging spring as the busiest time of year due to weeding and re-planting (Pilgrim et al. 2018). The emphasis on food forestry being a community project was clear in the solutions explored, such as working with volunteers (Pilgrim et al. 2018).

Secondly, revenue in food forestry comes from both produce and other diversified offerings. Albrecht and Wiek (2021a) observed schemes such as u-pick, community-supported agriculture, food boxes, and onsite and market sales. Social media is also seen as a valuable marketing and sales channel (Albrecht and Wiek 2021b; Yen and Chen 2016). Many food forests bring in revenue through education, consultancy, or other recreational events (Albrecht and Wiek 2021a, 2021b; Pilgrim et al. 2018; Hammersten et al. 2019). Overall economic viability is low for many of the food forests studied, presenting yet another research gap (Albrecht and Wiek 2021a; Pilgrim et al. 2018; Schafer et al. 2019).

While several studies have investigated success factors and challenges to initiating and establishing a food forest, there is room for considerably more research around practical aspects such as design, species assemblages, and ongoing management techniques.

Challenges and opportunities of food forestry

Opportunities

Food forests provide many opportunities. Based on the literature studied, we would like to highlight the opportunities for an urban environment and for Indigenous communities. In urban environments food forests can contribute to a city's goals around food security (Clark and Nicholas 2013), provide areas for leisure (Hong et al. 2020), function as a community hub (Passidomo 2016; Shi 2023), contribute to biodiversity (Thiesen et al. 2022), and reduce urban heat to mitigate climate change effects (Rockwell et al. 2022). Urban food forests are often located in close proximity to markets (Albrecht and Wiek 2021b). In addition to proximity to markets, their location also means they are close to communities, which creates the opportunity for community-based foraging (Clark and Nicholas 2013). Additionally, substantial areas of under-utilised public land—especially in low-income areas—could be transformed into edible urban infrastructure, like food forests, through policy reform and public–private partnerships (Hemmelgarn and Munsell 2021; Wiek and Albrecht 2022).

A significant number of studies highlight the value of incorporating Indigenous origins and knowledge in food forestry (Armstrong et al. 2023, 2021; Bosco and Thomas 2023; Hemmelgarn and Munsell 2021; Wartman et al. 2017; Yen and Chen 2016). Forest foods have historically been a vital component of many Indigenous foodways globally, including in temperate regions such as the Pacific Northwest of North America (e.g. Armstrong et al. 2023, 2021), where Indigenous forest food management practices created ecosystems with plant and functional trait diversity that persists, even 150 years after the management ceased. These systems are culturally significant, linking communities to their homelands and ancestral knowledge, and often serve as expressions of identity and self-determination (Bosco and Thomas 2023; Wartman et al. 2017). In Aotearoa New Zealand for example, this includes the incorporation and revitalisation of mahinga kai, a Māori system of food and resource management grounded in intergenerational knowledge, seasonal indicators, and reciprocal relationships with the environment (Smith and Hutchings 2025). Food is understood within a relational framework that connects people, ngā Atua (deities), land, and community, forming part of a wider Indigenous worldview. This interconnected approach offers pathways of hope for families and communities seeking alternatives to gaining back food sovereignty and identity.

Similar values are reflected in traditional Aboriginal land management systems in Australia. Bush tucker, a term used to describe native Australian foods traditionally harvested by Aboriginal for substance or medicinal use, reflects a deep body of ecological and Indigenous knowledge (Pascoe 2014). While not typically described using 'food forestry' in academic literature, First Nations scholars highlight practices such as seed propagation, harvest from flora and fauna, fire-stick farming, crop cultivation, and landscape scale harvesting were deliberately designed to sustain and enhance food resources (Clarke 2011). Similar to mahinga kai, these systems encompass broader frameworks of kinship, spiritual responsibility, Indigenous knowledge and ecological care. As with mahinga kai, these practices are being revitalised as part of efforts to restore cultural identity, and stewardship, and food sovereignty.

Incorporating Indigenous knowledge into food forestry offers not only ecological benefits but also cultural restoration, contributing to food sovereignty and reconnection with land. Studies highlight growing opportunities to restore Indigenous land and food practices through forest gardening, and how

to incorporate traditional Indigenous knowledge in today's food forestry systems (Bosco and Thomas 2023; Wartman et al. 2017). Building respectful partnerships and co-developing food forestry initiatives using culturally appropriate research methodology presents a significant opportunity for designing sustainable food forestry systems that support both ecological regeneration and community wellbeing.

Although temperate food forestry is presented as an alternative solution to growing healthy food, while also providing sustainable livelihoods and environmental services (Albrecht and Wiek 2021a), few papers provide evidence for the environmental benefits of temperate food forestry. Research reported by Rockwell (2022) about species richness and ecosystem services in the southeastern part of the United States of America finds that food forestry can contribute to urban greening. Three papers discuss soil carbon storage under food forests in the United Kingdom, and find that although these forests are primarily designed for food provision, they can function as a valuable carbon sink (Hald and Schafer 2023; Lehmann et al. 2019; Schafer et al. 2019). Russo et al. (2017) highlight the fact that food forestry can contribute to social, environmental, and economic sustainability, including food security. However, trade-offs, such as potential health risks due to exposure to heavy metals and organic chemical contaminants that can be present in urban areas, need to be considered. Although food forestry is mentioned as a potential food production solution, the relatively low number of papers focusing on evidence for environmental benefits of temperate food forestry indicates a gap in the literature, with no studies specifically focused on biodiversity effects or water quality impacts.

Challenges

Despite the opportunities presented above, food forests also come with a variety of challenges, which are worth noting. These include logistics, health and safety, land access, and governance. Logistical challenges are pertinent around harvesting. Castro et al. (2018) reviewed research on edible urban forests and noted the need for food processing or sufficient demand when the edible species were ready to avoid unharvested fruit attracting pests or becoming unsightly. Kowalski and Conway (2023) interviewed municipal officials and found basic knowledge gaps about how to care for urban fruit trees. Issues like these may be overcome through public–private partnerships such as those observed in Beacon Food Forest where the city of Seattle brings in the expertise of volunteers and community groups (Albrecht and Wiek 2021b; Wiek and Albrecht 2022).

Urban food safety is another critical area requiring further research, including distances from sources of pollution and barriers that can reduce airborne pollution entering food forest soil. A greater understanding of the ways heavy metals and other pollutants accumulate in popular temperate food forest plants would also support knowledge of which plants, and parts of plants, are safe to eat when grown with different levels of soil contamination (Albrecht and Wiek 2021b; Castro et al. 2018; Hald and Schafer 2023). Incorporating phytoremediation—using plants to 'clean up' contaminated environments—into food forestry design could help improve both ecological function and food safety in urban contexts.

An additional challenge for Indigenous communities is land access and governance. For example, in Aotearoa New Zealand, Māori land tenure complexities and loss of access to traditional food gathering sites can limit the establishment of food forests on traditionally held lands (Smith and Hutchings 2025). In Canada, First Nations face similar challenges, particularly in urban contexts, where relationships with landowners, control over land, and governance systems shape access (Miltenburg et al. 2023). Similar governance constraints around access to land are evident in urban contexts. Kowalski and Conway (2023) for example found that while interest in urban food trees is growing, most local councils provide only limited support and lack clear policies that enable harvesting or long-term community stewardship. In many cases, local by-laws prohibit the removal of plant material, directly conflicting with the purpose of food forests. The literature highlights a broader need for co-governance models and policy shifts that support inclusive, community-led management of food forests in public spaces (Kowalski and Conway 2023; Smith and Hutchings 2025). Reconnecting with ancestral food landscapes is essential not only for restoring food security, but also for enabling self-determination through Indigenous food systems (Smith and Hutchings 2025). Addressing these

challenges requires structural support for Indigenous governance of land.

Temperate food forestry: a research agenda

After reviewing the existing literature we identified four areas for further research: i) more studies in temperate areas in the southern hemisphere, such as, Oceania, Southern Africa, and South America: how can these be increased?; ii) Integrating Indigenous knowledge into food forestry: how can this be done?; iii) Economic viability: what is the potential revenue model, especially for rural food forests?; and iv) Mature food forests: how do we manage these?

Research topic 1: southern hemisphere-specific studies

Despite growing interest in food forestry globally, peer-reviewed research from Oceania, Southern Africa, and South America, remains limited. Of the 44 studies reviewed, we found no examples of food forests from Australia, South Africa, or South America, and only one study from Aotearoa New Zealand. While relevant practices may exist in these regions, particularly within Indigenous systems, they are often described using different terms and we note this could be a case of the use of particular language, and the fact that the naming of 'food forests' and 'food forestry' is a relatively new term. Regardless, this points to a major gap in understanding how food forestry could function across the diverse ecological and cultural landscapes of these southern temperate regions, as well as the ways in which Indigenous Peoples from these countries have been cultivating, managing, and amplifying forest foods and resources over the centuries. Further research should identify relevant native and introduced species best suited to temperate food forest systems, including species combinations, ecological functions, seasonal dynamics, and cultural relevance to regional climates. Key research questions include: 'How can native species be incorporated into temperate food forestry in the Southern Hemisphere?'; and should link up with topic two, by focusing on the question 'How can culturally relevant species be included in temperate food forestry?'.

Research topic 2: integrating Indigenous knowledge and food forestry

Indigenous knowledge offers valuable insights into sustainable land management, biodiversity, and long-term ecosystem health, making it essential to the development of resilient food forestry systems. Literature across multiple contexts, including Aotearoa New Zealand, Taiwan, and North America, emphasises the importance of learning from Indigenous foodways and restoring cultural food systems (Armstrong et al. 2023, 2021; Bosco and Thomas 2023; Yen and Chen 2016). This includes understanding the cultural, spiritual, and seasonal dimensions of food cultivation. It is vital to continue to expand this field of research, including further archaeological surveys and partnerships with communities seeking to revitalise traditional practices. The way this research is done is just as important as the results reported, with approaches like values-led, participatory action, and decolonial and reconciliation-based research recommended in the literature (Bosco and Thomas 2023). Key questions to address are: 'How can Indigenous governance and land relationships be embedded into food forest design?'; 'How can food forests be designed and managed in a way that enhances cultural identity and supports the restoration of food sovereignty?'; and, 'How can Indigenous-led models be scaled in both urban and rural settings in ways that actively engage and are driven by the community?'.

Research topic 3: economic viability

Food forestry enterprises show success in individual, family, and community settings. As reported by Clark and Nicholas (2013), large-scale production generally requires a major investment in planning and staff resource and capability. As a result, food forests were mostly created as community projects by nonprofit organisations or as private–public partnerships—leaning towards social or environmental outcomes rather than economic ones (Albrecht and Wiek 2021a; Pilgrim et al. 2018). Hence, a significant gap in the literature pertaining to the economic viability of medium- and large-scale food forests exists, specifically around examining both its contributions to the economic viability of rural enterprises and the ability to scale these (Muscas et al. 2024). Any valuations should go beyond a purely monetary value though, by

including a variety of ecosystem services that might benefit from the efforts, such as biodiversity, carbon sequestration, erosion reduction, and recreation. Research questions to answer include: 'What ecosystem services does food forestry provide (and can these be quantified)?'; 'How should rural food forests be designed to allow efficient, low-cost harvesting practices?'; 'What potential yields can a food forest provide in a way that is commercially viable?'; 'What types of products can provide a consistent revenue stream for food forests?'; and, 'What workforce composition is needed to run a large-scale food forest?'.

Research topic 4: mature food forests

Literature has shown that most studies on temperate food forestry have focused on recently established forests. This focus is understandable, as temperature food forestry is a relatively new area of research. With many food forests moving beyond the establishment phase, we see the benefit of more research into these forests. Further research is needed to answer questions such as: 'How do we maintain established food forests?'; 'How can sustained harvesting plans be set up?'; and connect to research topic three by answering the question: 'What revenue models are available and how do we work towards these?'.

Conclusions

Temperate food forestry shows some promise at the individual, family, and community level—particularly so in the more densely-settled Northern Hemisphere—and is becoming an established area of research. However, several gaps remain before objective evidence can conclude the exact level of promise. The aim of this research was to: (i) synthesise the current state of the literature; (ii) determine what this literature tell us about the main research themes in the temperate food forestry literature; and (iii) identify knowledge tensions and gaps to direct future research into this area. Although a relatively young research area, food forests in temperate climates show some promise—not just environmentally and socially, but also, economically. But more research is required into temperate food forestry in the Southern Hemisphere, the incorporation of Indigenous knowledge in these systems, the economics of these forests, and the running of mature food forests.

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Author's contribution JK, CO, HT co-coordinated the project J.K., C.O., H.T. and B.P. did the initial search for the articles and selected the articles relevant for review. B.P. conducted the initial read through all the articles included in the review. JK, CO, HT reviewed the initial read of the articles BP wrote summaries of each of the articles and a first draft of the manuscript JK, CO, HT reviewed and revised the manuscript JK coordinated the submission of the manuscript to the journal.

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Declarations

Conflict of interest The authors have no competing interests to declare.

Appendix A –An overview of the 43 studies included in the literature review

References	Title	Journal	Focus: Urban / Rural	Focus: Social (S), Economic (E), Biophysical (B)
Albrecht and Wiek (2021a)	Implementing sustainable food forests: extracting success factors through a cross—case comparison	Journal of Agriculture, Food Systems and Community Development	Rural	All

References	Title	Journal	Focus: Urban / Rural	Focus: Social (S), Economic (E), Biophysical (B)
Albrecht and Wiek (2021b)	Food forests: Their services and sustainability	Journal Of Agriculture Food Systems And Community Development	Urban	All
Armstrong et al. (2023)	Historical ecology of forest garden management in Laxyuubm Ts'msyen and beyond	Ecosystems and People	Urban	S
Armstrong et al. (2021)	Historical Indigenous Land-Use Explains Plant Functional Trait Diversity	Ecology and Society	Urban	S, B
Askerlund and Almers (2016)	Forest gardens—new opportunities for urban children to understand and develop relationships with other organisms	Urban Forestry & Urban Greening	Rural	S, E
Bellows et al. (2023)	Foraging Wild Edibles: Dietary Diversity in Expanded Food Systems	Nutrients	Urban	B
Bjorklund et al. (2019)	Exploring the potential of edible forest gardens: experiences from a participatory action research project in Sweden	Agroforestry Systems	Rural	B
Bosco and Thomas (2023)	The Skarù·rę? (Tuscarora) Food Forest Project—Reconciliation in Sustainable Agriculture Research and Education through Cross-Cultural Agroforestry Demonstration	Ethnobiology Letters	Urban	S, E
Carinanos et al. (2022)	The Place of Urban Food Forests in Cities of the 21st Century	Urban Planning	Rural	S, E
Castro et al. (2018)	"Edible" urban forests as part of inclusive, sustainable cities	Unasylva (English ed.)	Urban	All

References	Title	Journal	Focus: Urban / Rural	Focus: Social (S), Economic (E), Biophysical (B)
Clark and Nicholas (2013)	Introducing urban food forestry: a multi-functional approach to increase food security and provide ecosystem services	Landscape Ecology	Rural	S, E
Hald and Schafer (2023)	Temperate food forest soil and dead organic matter carbon content relative to adjacent land	Agrosystems Geosciences & Environment	Urban	S
Hammarsten et al. (2019)	Developing ecological literacy in a forest garden: children's perspectives	Journal Of Adventure Education And Outdoor Learning	Rural	All
Hemmelgarn and Munsell (2021)	Exploring 'beyond-food' opportunities for biocultural conservation in urban forest gardens	Urban Agriculture & Regional Food Systems	Rural	All
Hong et al. (2020)	Development of a Evaluation Criteria of the Forest Garden	The International Journal of Advanced Culture Technology	Urban	S
Kowalski and Conway (2019)	Branching out: The inclusion of urban food trees in Canadian urban forest management plans	Urban Forestry & Urban Greening	Urban	S, E
Kowalski and Conway (2023)	The routes to fruit: Governance of urban food trees in Canada	Urban Forestry & Urban Greening	Urban	B
Lehmann et al. (2019)	Quantification of the understorey contribution to carbon storage in a peri-urban temperate food forest	Urban Forestry & Urban Greening	Urban	B
Lovell et al. (2021)	Community Orchards for Food Sovereignty, Human Health, and Climate Resilience: Indigenous Roots and Contemporary Applications	Forests	Urban	All

References	Title	Journal	Focus: Urban / Rural	Focus: Social (S), Economic (E), Biophysical (B)
Muscas et al. (2024)	Potential Economic and Ecosystem Performances of Some Mediterranean Fruit Plants in an Urban Context	Sustainability	Urban	All
Nytofte et al. (2019)	Sustainable food production in a temperate climate—a case study analysis of the nutritional yield in a peri-urban food forest	Urban Forestry & Urban Greening	Urban	B
Oncini et al. (2024)	Where the wild things are: How urban foraging and food forests can contribute to sustainable cities in the Global North	Urban Forestry & Urban Greening	Rural	S, E
Park and Higgs (2018)	A criteria and indicators monitoring framework for food forestry embedded in the principles of ecological restoration	Environmental Monitoring And Assessment	Urban	S, B
Park et al. (2019)	Urban food systems that involve trees in Northern America and Europe: A scoping review	Urban Forestry & Urban Greening	Rural	S, E
Park and Higgs (2018)	Exploring the potential of food forestry to assist in ecological restoration in North America and beyond	Restoration Ecology	N/A	All
Passidomo (2016)	Community gardening and governance over urban nature in New Orleans's Lower Ninth Ward	Urban Forestry & Urban Greening	Rural	B

References	Title	Journal	Focus: Urban / Rural	Focus: Social (S), Economic (E), Biophysical (B)
Pilgrim et al. (2018)	Evaluating the multiple benefits of multi-layered agroforestry systems	International Journal of Agricultural Management	Urban	All
Riolo (2019)	The social and environmental value of public urban food forests: The case study of the Picasso Food Forest in Parma, Italy	Urban Forestry & Urban Greening	Urban	S, B
Rockwell et al. (2022)	Species Richness, Stem Density, and Canopy in Food Forests: Contributions to Ecosystem Services in an Urban Environment	Urban Planning	Urban	S, B
Roodhof (2024)	Understanding the emerging phenomenon of food forestry in the Netherlands: An assemblage theory approach	Journal Of Agriculture Food Systems And Community Development	Urban	B
Roodhof and Veen (2021)	Finding quality in quantitative methods: the case of food forestry in the Netherlands	Urban Agriculture and Regional Food Systems	Urban	S
Russo et al. (2017)	Edible green infrastructure: An approach and review of provisioning ecosystem services and disservices in urban environments	Agriculture Ecosystems & Environment	Urban	All
Schafer et al. (2019)	Tree layer carbon stock quantification in a temperate food forest: A peri-urban polyculture case study	Urban Forestry & Urban Greening	Urban	S, E
Schaffer et al. (2019)	Can Agroforestry Grow beyond Its Niche and Contribute to a Transition towards Sustainable Agriculture in Sweden?	Sustainability	Rural	B

References	Title	Journal	Focus: Urban / Rural	Focus: Social (S), Economic (E), Biophysical (B)
Shi (2023)	The urban food forest: Creating a public edible landscape	Urban Design International	Urban	B
Smith and Hutchings (2024)	Feeding Indigenous Aotearoa better	Journal of the Royal Society of New Zealand	Rural	S
Stoltz and Schaffer (2018)	Salutogenic Affordances and Sustainability: Multiple Benefits With Edible Forest Gardens in Urban Green Spaces	Frontiers In Psychology	Rural	B
Thiesen et al. (2022)	An Ecosystem Service Approach to Assessing Agro-Ecosystems in Urban Landscapes	Land	Urban	S, B
Wartman et al. (2017)	The establishment of apple orchards as temperate forest garden systems and their impact on indigenous bacterial and fungal population abundance in Southern Ontario, Canada	Renewable Agriculture And Food Systems	Urban	B
Wartman et al. (2018)	Temperate Agroforestry: How Forest Garden Systems Combined with People-Based Ethics Can Transform Culture	Sustainability	Urban	S, E
Wiek and Albrecht (2022)	Almost there: on the importance of a comprehensive entrepreneurial ecosystem for developing sustainable urban food forest enterprises	Urban Agriculture & Regional Food Systems	Urban	S, E

References	Title	Journal	Focus: Urban / Rural	Focus: Social (S), Economic (E), Biophysical (B)
Wiersum (2004)	Forest gardens as an 'intermediate' land-use system in the nature-culture continuum: Characteristics and future potential	Agroforestry Systems	Rural	S, E
Yen and Chen (2016)	Agroforestry as sustainable agriculture: An observation of tayal indigenous people's collective action in Taiwan	International Journal of Environmental Sustainability	Rural	S

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